

Arya Mohammadi

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EDUCATION

University of California - San Diego

B.S. in Mathematics and Computer Science, GPA: 3.78/4.00

Expected June 2027

Dean's List (4x)

Coursework: Data Structures, Algorithms, Software Construction, Systems Programming, Discrete Structures, Linear Algebra, Probability & Statistics, Assembly Language

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL, HTML

Frameworks & Libraries: Next.js, React.js, Node.js, Express.js, Mongoose, OpenAI API, Zustand, Jest, Supertest, GoogleTest, Playwright

Tools & Platforms: Git, Linux, MongoDB, Cloudinary, Twilio, Vercel, Railway, CMake, gdb, Valgrind

Concepts: REST APIs, AI Integration, Authentication, Full-Stack Development, Unit Testing, CI/CD

EXPERIENCE

Computer Science and Mathematics Tutor

September 2024 – June 2025

University of California, Riverside

- Tutored 400+ students over 600 hours in DSA and systems programming, improving average grade from B- to A-.
- Debugged 100+ C++ codebases, reducing average student debugging time by 40%.
- Created 50+ practice problems for programming foundations and time complexity; adopted by 3 tutors.

Hardware Intern

June 2021 – September 2023

ProCustom Group

- Assembled 250+ rugged laptop systems for Lockheed Martin and Raytheon with 100% on-time delivery.
- Fabricated 160+ custom cables and applied EMI shielding, achieving 100% military standard compliance.
- Documented 50+ production tasks for quality control, contributing to AS9100D certification maintenance.

PROJECTS

CodeLens | *Next.js, TypeScript, OpenAI API, Zustand, Tailwind CSS, Vercel*

- Built AI-powered code review tool that detects bugs, suggests optimizations, and analyzes time/space complexity across 8 programming languages using GPT-4.
- Designed REST API with structured JSON schema for consistent AI responses; implemented Zustand for client-side state management with async error handling.
- Added syntax highlighting, skeleton loaders, and staggered animations; deployed on Vercel edge network.

DormDuos.com | *React.js, Node.js, Express.js, MongoDB, JWT, Jest, Vercel, Railway*

- Built student housing platform matching UCR renters with listings; implemented React frontend, Node.js/Express API, JWT auth, Cloudinary uploads, and input sanitization.
- Reduced query latency to <1ms by adding compound indexes on status, price, and room filters.
- Benchmarked API at 28k req/sec with Autocannon; stress-tested search and pagination with 10k+ listings.
- Wrote tests with 100% branch coverage using Jest, Supertest, and MongoDB Memory Server.

AryaElectric.com | *Next.js, Node.js, Twilio, Nodemailer, Jest, Playwright*

- Built production website for electrical contractor with async API routes for quote submissions, SMS webhooks, and automated follow-ups.
- Added server-side caching for Google Reviews API, reducing cached response latency by 90%.
- Wrote 90+ tests with mocked Twilio/Nodemailer; validated end-to-end flows with Playwright.

Battlestar RPG | *C++, STL, GoogleTest, CMake*

- Built 3.4k-line RPG engine with 23 classes over 3 months in a team of 5.
- Implemented MaxHeap-based combat scheduler for $O(n \log n)$ speed-based turn ordering.
- Designed polymorphic item system with clone pattern and template-based save/load serialization.
- Wrote 2.3k lines of unit tests covering combat, inventory, sorting, and state persistence.

Chat Server | *C, HTTP, Sockets, Linux*

- Built a multi-client chat server in C with custom HTTP parsing, supporting new posts, message rendering, and reactions.
- Designed in-memory data models for chats and reactions with size limits, timestamps, and UTF-8-safe fields.
- Streamed formatted chat output using `write()` for efficient network responses.